RAMAS ACADEMY

Class 12 · Board Exam Preparation · OMR Test Series

Class 12 Chemistry – OMR Chapter Test

NCERT Chemistry Part I (2026–27) · Unit 5: Coordination Compounds · based on CBSE board questions (2020–2025)

Time Allowed: 3 Hours | **Total Questions: 60** | **Maximum Marks: 80**

General Instructions

1. This paper has **60 multiple-choice questions** in five sections (A–E). Every question has **exactly one** correct option.
2. Mark your answers **only on the OMR answer sheet** by completely darkening one bubble with a **black/blue ball-point pen**. Do not use pencil, whitener or correction fluid.
3. Marks differ by section: see the table below. **Negative marking: 25% of the question's marks** is deducted for each wrong answer. Darkening more than one bubble counts as a wrong answer. Unattempted questions get 0.
4. Section D has four case studies with four MCQs each. Read the passage before answering.
5. Questions in Sections C and E need full working on rough paper; the options are designed so that guessing is costly. Use the separate rough sheets provided; nothing written on the question paper is checked.
6. Calculators and other electronic devices are not allowed. Atomic numbers: Ti 22, Cr 24, Mn 25, Fe 26, Co 27, Ni 28, Cu 29, Zn 30, Pt 78. Spin-only magnetic moment $\mu = \sqrt{[n(n+2)]}$ BM. Questions are taken from or closely modelled on CBSE board papers of the last six years.

Section	Type	Questions	No.	Correct	Wrong	Marks
A	1-mark level (incl. Assertion–Reason)	Q1–33	33	+1	−0.25	33
B	2-mark level (Very Short Answer type)	Q34–39	6	+2	−0.5	12
C	3-mark level (Short Answer type)	Q40–42	3	+3	−0.75	9
D	4-mark level (Case Studies – 4 MCQs per case)	Q43–58	16	+1	−0.25	16
E	5-mark level (Long Answer type)	Q59–60	2	+5	−1.25	10
Total			**60**	**Unattempted: 0**		**80**

Section A

1-mark level (incl. Assertion–Reason) · 33 × 1 = 33 marks · −0.25 per wrong

Q1. According to Werner's theory, the primary valency of a metal ion in a complex corresponds to its: Werner's Theory & Key Terms [1]
(A) oxidation state
(B) number of ligands
(C) number of unpaired electrons
(D) coordination number

Q2. One mole of $CoCl_3 \cdot 4NH_3$ reacts with excess $AgNO_3$ to give 1 mol of AgCl. Its formula is: Werner's Theory & Key Terms [1]
(A) $[Co(NH_3)_4]Cl_3$
(B) $[Co(NH_3)_4Cl]Cl_2$
(C) $[Co(NH_3)_4Cl_2]Cl$
(D) $[Co(NH_3)_3Cl_3] \cdot NH_3$

Q3. The coordination number and oxidation state of Fe in $[Fe(C_2O_4)_3]^{3-}$ are: Werner's Theory & Key Terms [1]
(A) 3 and +6 **(B)** 3 and +3
(C) 6 and +2 **(D)** 6 and +3

Q4. $EDTA^{4-}$ is a: Werner's Theory & Key Terms [1]
(A) tetradentate ligand **(B)** hexadentate ligand
(C) monodentate ligand **(D)** bidentate ligand

Q5. Which of the following is an ambidentate ligand? Werner's Theory & Key Terms [1]
(A) NO_2^- **(B)** NH_3
(C) H_2O **(D)** en

Q6. The number of ions produced in solution by one formula unit of $[Co(NH_3)_6]Cl_3$ is: Werner's Theory & Key Terms [1]
(A) 3 **(B)** 9 **(C)** 4 **(D)** 6

Q7. A ligand that binds to a metal through two donor atoms forming a ring is called a: Werner's Theory & Key Terms [1]
(A) monodentate ligand **(B)** chelating ligand
(C) ambidentate ligand **(D)** bridging ligand only

Q8. **Assertion (A):** Chelate complexes are more stable than similar complexes with monodentate ligands.
Reason (R): Chelating ligands form rings with the metal atom (chelate effect). Werner's Theory & Key Terms [1]
(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

Q9. The IUPAC name of $[Co(NH_3)_5Cl]Cl_2$ is: IUPAC Nomenclature [1]
(A) pentaamminechloridocobaltate(III) chloride
(B) pentaamminechloridocobalt(II) chloride
(C) chloridopentaamminecobalt(III) chloride
(D) pentaamminechloridocobalt(III) chloride

Q10. The IUPAC name of $K_3[Fe(CN)_6]$ is: IUPAC Nomenclature [1]
(A) tripotassium hexacyanoiron(III)
(B) potassium hexacyanidoiron(III)
(C) potassium hexacyanidoferrate(II)
(D) potassium hexacyanidoferrate(III)

Q11. The formula of tetraamminediaquacobalt(III) chloride is: IUPAC Nomenclature [1]
(A) $[Co(NH_3)_4(H_2O)_2]Cl$
(B) $[Co(NH_3)_4(H_2O)_2]Cl_3$
(C) $[Co(NH_3)_4(H_2O)_2]Cl_2$
(D) $[Co(NH_3)_4(H_2O)Cl]Cl_2$

Q12. The IUPAC name of $[Pt(NH_3)_2Cl_2]$ is: IUPAC Nomenclature [1]
(A) dichloridodiammineplatinum(II)
(B) diamminedichloridoplatinum(IV)
(C) diamminedichloridoplatinum(II)
(D) diamminedichloridoplatinate(II)

Q13. The oxidation state of nickel in $[Ni(CO)_4]$ is: IUPAC Nomenclature [1]
(A) +2 **(B)** −2 **(C)** +4 **(D)** 0

Q14. The IUPAC name of $K_2[Zn(OH)_4]$ is: IUPAC Nomenclature [1]
(A) potassium tetrahydroxidozincate(IV)
(B) dipotassium tetrahydroxozinc(IV)
(C) potassium tetrahydroxidozincate(II)
(D) potassium tetrahydroxidozinc(II)

Q15. The IUPAC name of $[Cr(en)_3]Cl_3$ is: IUPAC Nomenclature [1]
(A) tri(ethylenediamine)chromate(III) chloride
(B) tris(ethane-1,2-diamine)chromium(III) chloride
(C) triethylenediaminechromium(III) trichloride
(D) tris(ethane-1,2-diamine)chromium(II) chloride

Q16. In the IUPAC name of a complex anion, the name of the central metal ends in: IUPAC Nomenclature [1]
(A) -ate **(B)** -ide **(C)** -ite **(D)** -ium

Q17. **Assertion (A):** The IUPAC name of $[Ni(CN)_4]^{2-}$ is tetracyanidonickelate(II) ion.
Reason (R): In an anionic complex, the name of the metal ends with the suffix '-ate'. IUPAC Nomenclature [1]
(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

Q18. $[Co(NH_3)_5(NO_2)]Cl_2$ and $[Co(NH_3)_5(ONO)]Cl_2$ show: Isomerism [1]
(A) geometrical isomerism **(B)** coordination isomerism
(C) linkage isomerism **(D)** ionisation isomerism

Q19. $[Co(NH_3)_5SO_4]Br$ and $[Co(NH_3)_5Br]SO_4$ are examples of: [1]
Isomerism
(A) hydrate isomers **(B)** ionisation isomers
(C) optical isomers **(D)** linkage isomers

Q20. $[Cr(H_2O)_6]Cl_3$ (violet) and $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$ (grey-green) show: Isomerism [1]
(A) coordination isomerism
(B) linkage isomerism
(C) optical isomerism
(D) solvate (hydrate) isomerism

Q21. $[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$ are: Isomerism [1]
(A) ionisation isomers **(B)** geometrical isomers
(C) coordination isomers **(D)** linkage isomers

Q22. A square planar complex $[MA_2B_2]$ can show: Isomerism [1]
(A) both geometrical and optical isomerism
(B) no isomerism
(C) optical isomerism
(D) geometrical (cis–trans) isomerism

Q23. The number of geometrical isomers of $[Co(NH_3)_3(NO_2)_3]$ (octahedral MA_3B_3) is: Isomerism [1]
(A) 2 (fac and mer) **(B)** none
(C) 3 **(D)** 4

Q24. Which of the following complexes is optically active? Isomerism [1]
(A) trans-$[Pt(NH_3)_2Cl_2]$
(B) cis-$[Co(en)_2Cl_2]^+$
(C) trans-$[Co(en)_2Cl_2]^+$
(D) $[Co(NH_3)_6]^{3+}$

Q25. **Assertion (A):** Tetrahedral complexes do not show geometrical isomerism. [1]
Reason (R): In a tetrahedral complex, the relative positions of the ligands attached to the central atom are the same with respect to each other. Isomerism
(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

Q26. The hybridisation and magnetic nature of $[Ni(CN)_4]^{2-}$ are: [1]
Bonding (VBT & CFT), Colour & Magnetism
(A) dsp^2, paramagnetic **(B)** dsp^2, diamagnetic
(C) sp^3, paramagnetic **(D)** sp^3, diamagnetic

Q27. The hybridisation and shape of $[NiCl_4]^{2-}$ are: [1]
Bonding (VBT & CFT), Colour & Magnetism
(A) dsp^2, square planar **(B)** d^2sp^3, octahedral
(C) sp^3, tetrahedral **(D)** sp^3d^2, octahedral

Q28. $[Co(NH_3)_6]^{3+}$ is an inner orbital complex while $[CoF_6]^{3-}$ is an outer orbital complex. The number of unpaired electrons in $[CoF_6]^{3-}$ is: Bonding (VBT & CFT), Colour & Magnetism [1]
(A) 2 **(B)** 0 **(C)** 3 **(D)** 4

Q29. In an octahedral crystal field, the d orbitals split into: [1]
Bonding (VBT & CFT), Colour & Magnetism
(A) lower e_g (two) and higher t_{2g} (three)
(B) five degenerate orbitals
(C) lower e (two) and higher t_2 (three)
(D) lower t_{2g} (three) and higher e_g (two)

Q30. Which ligand produces the strongest crystal field splitting? [1]
Bonding (VBT & CFT), Colour & Magnetism
(A) CN^- **(B)** H_2O **(C)** F^- **(D)** Cl^-

Q31. The spin-only magnetic moment of $[Fe(CN)_6]^{3-}$ is about: [1]
Bonding (VBT & CFT), Colour & Magnetism
(A) 5.92 BM **(B)** 1.73 BM **(C)** 0 BM **(D)** 2.83 BM

Q32. Anhydrous $CuSO_4$ is white but $CuSO_4 \cdot 5H_2O$ is blue because: [1]
Bonding (VBT & CFT), Colour & Magnetism
(A) the anhydrous salt absorbs all visible light
(B) anhydrous $CuSO_4$ has no d electrons
(C) in the anhydrous salt there are no water ligands to cause d–d splitting
(D) water reduces Cu^{2+} to Cu^+

Q33. **Assertion (A):** $[Ti(H_2O)_6]^{3+}$ is coloured. [1]
Reason (R): Ti^{3+} has one d electron that can undergo a d–d transition by absorbing visible light.
Bonding (VBT & CFT), Colour & Magnetism
(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

Section B
2-mark level (Very Short Answer type) · 6 × 2 = 12 marks · −0.5 per wrong

Q34. When 0.1 mol of $CoCl_3(NH_3)_5$ is treated with excess $AgNO_3$, 0.2 mol of AgCl is obtained. The conductivity of its solution will correspond to: Werner's Theory & Key Terms [2]
(A) a 1 : 1 electrolyte **(B)** a 1 : 3 electrolyte
(C) a 1 : 2 electrolyte **(D)** a non-electrolyte

Q35. The correct formula of potassium tri(oxalato)chromate(III) and of hexaamminecobalt(III) sulphate are: IUPAC Nomenclature [2]
(A) $K_2[Cr(C_2O_4)_3]$ and $[Co(NH_3)_6]SO_4$
(B) $K[Cr(C_2O_4)_3]$ and $[Co(NH_3)_6]_3(SO_4)_2$
(C) $K_3[Cr(C_2O_4)_3]$ and $[Co(NH_3)_6](SO_4)_3$
(D) $K_3[Cr(C_2O_4)_3]$ and $[Co(NH_3)_6]_2(SO_4)_3$

Q36. The total number of isomers (geometrical + optical) of $[Co(en)_2Cl_2]^+$ is: Isomerism [2]
(A) 4 **(B)** 3 **(C)** 1 **(D)** 2

Q37. How many geometrical isomers are possible for the square planar complex $[Pt(NH_3)(Br)(Cl)(py)]$? Isomerism [2]
(A) 0 **(B)** 4 **(C)** 3 **(D)** 2

Q38. Using CFT, the electronic configuration of d^4 ion in an octahedral field when $\Delta_o > P$ (pairing energy) is: [2]
Bonding (VBT & CFT), Colour & Magnetism
(A) $t_{2g}^4 e_g^0$ **(B)** $e_g^4 t_{2g}^0$
(C) $t_{2g}^3 e_g^1$ **(D)** $t_{2g}^2 e_g^2$

Q39. $[Fe(H_2O)_6]^{3+}$ is strongly paramagnetic whereas $[Fe(CN)_6]^{3-}$ is weakly paramagnetic. The numbers of unpaired electrons are respectively: Bonding (VBT & CFT), Colour & Magnetism [2]
(A) 1 and 5 **(B)** 5 and 1 **(C)** 3 and 1 **(D)** 5 and 3

Section C
3-mark level (Short Answer type) · 3 × 3 = 9 marks · −0.75 per wrong

Q40. Write the IUPAC names of: (i) $[Co(NH_3)_4(H_2O)Cl]Cl_2$, (ii) $[Ni(CO)_4]$, (iii) $K_3[Cr(C_2O_4)_3]$. The correct set is: IUPAC Nomenclature [3]
(A) (i) tetraammineaquachloridocobalt(III) chloride; (ii) nickel tetracarbonyl(II); (iii) tripotassium chromium oxalate
(B) (i) aquatetraamminechloridocobalt(III) chloride; (ii) tetracarbonylnickelate(0); (iii) potassium trioxalatochromate(III)
(C) (i) tetraammineaquachloridocobalt(II) chloride; (ii) tetracarbonylnickel(II); (iii) potassium trioxalatochromium(III)
(D) (i) tetraammineaquachloridocobalt(III) chloride; (ii) tetracarbonylnickel(0); (iii) potassium trioxalatochromate(III)

Q41. Using valence bond theory, predict the hybridisation, shape and magnetic behaviour of $[Co(NH_3)_6]^{3+}$ (Co: Z = 27). [3]
Bonding (VBT & CFT), Colour & Magnetism
(A) sp^3, tetrahedral, diamagnetic
(B) d^2sp^3, octahedral, paramagnetic
(C) sp^3d^2, octahedral, paramagnetic
(D) d^2sp^3, octahedral, diamagnetic

Q42. Calculate the crystal field stabilisation energy (CFSE) of a d^6 ion in a strong-field (low-spin) octahedral complex, ignoring pairing energy. [3]
Bonding (VBT & CFT), Colour & Magnetism

(A) 0 **(B)** $-0.4\ \Delta_o$
(C) $-2.4\ \Delta_o$ **(D)** $-1.6\ \Delta_o$

Section D

4-mark level (Case Studies – 4 MCQs per case) · 16 × 1 = 16 marks · −0.25 per wrong

Case Study 1: Werner's Cobalt Ammines

Werner's Theory & Key Terms

Werner prepared complexes of composition $CoCl_3{\cdot}6NH_3$ (yellow), $CoCl_3{\cdot}5NH_3$ (purple) and $CoCl_3{\cdot}4NH_3$ (green and violet forms). He treated each with excess $AgNO_3$ solution.

Answer Q43–46 on the basis of the above information.

Q43. The moles of AgCl precipitated per mole of $CoCl_3{\cdot}6NH_3$ are: [1]
(A) 0 **(B)** 3 **(C)** 1 **(D)** 2

Q44. The formula of the purple complex is: [1]
(A) $[Co(NH_3)_5Cl]Cl_2$ **(B)** $[Co(NH_3)_6]Cl_3$
(C) $[Co(NH_3)_5]Cl_3$ **(D)** $[Co(NH_3)_5Cl_2]Cl$

Q45. The green and violet forms of $CoCl_3{\cdot}4NH_3$ are: [1]
(A) trans and cis geometrical isomers
(B) coordination isomers
(C) linkage isomers
(D) ionisation isomers

Q46. The secondary valency of cobalt in all these complexes is: [1]
(A) 4 **(B)** 3 **(C)** 5 **(D)** 6

Case Study 2: Cisplatin

Isomerism

cis-$[Pt(NH_3)_2Cl_2]$ (cisplatin) is an anticancer drug; its trans isomer is not effective.

Answer Q47–50 on the basis of the above information.

Q47. The geometry of the complex is: [1]
(A) octahedral **(B)** square planar
(C) tetrahedral **(D)** trigonal bipyramidal

Q48. The oxidation state of Pt is: [1]
(A) 0 **(B)** +4 **(C)** +2 **(D)** +3

Q49. The type of isomerism shown by $[Pt(NH_3)_2Cl_2]$ is: [1]
(A) linkage **(B)** optical
(C) geometrical **(D)** ionisation

Q50. Cisplatin is optically: [1]
(A) inactive, because it is tetrahedral
(B) inactive, because it has a plane of symmetry
(C) active, because Pt is chiral
(D) active, because it is cis

Case Study 3: Colours of Nickel Complexes

Bonding (VBT & CFT), Colour & Magnetism

$[Ni(H_2O)_6]^{2+}$ is green, $[Ni(NH_3)_6]^{2+}$ is blue-violet and $[Ni(CN)_4]^{2-}$ is colourless/yellow. Ni: Z = 28.

Answer Q51–54 on the basis of the above information.

Q51. The number of unpaired electrons in $[Ni(H_2O)_6]^{2+}$ is: [1]
(A) 0 **(B)** 1 **(C)** 4 **(D)** 2

Q52. The spin-only magnetic moment of $[Ni(NH_3)_6]^{2+}$ is: [1]
(A) 2.83 BM **(B)** 4.90 BM **(C)** 1.73 BM **(D)** 0 BM

Q53. Replacing H_2O by NH_3 shifts the absorbed light to a shorter wavelength because NH_3: [1]
(A) is a stronger field ligand, increasing Δ_o
(B) is a weaker field ligand
(C) reduces Ni^{2+}
(D) has no lone pair

Q54. $[Ni(CN)_4]^{2-}$ is: [1]
(A) square planar and paramagnetic
(B) octahedral and diamagnetic
(C) tetrahedral and paramagnetic
(D) square planar and diamagnetic

Case Study 4: Crystal Field in Iron Complexes

Bonding (VBT & CFT), Colour & Magnetism

The pairing energy P for Fe^{2+} is about 210 kJ mol^{-1}. Δ_o for $[Fe(H_2O)_6]^{2+}$ is about 125 kJ mol^{-1} and for $[Fe(CN)_6]^{4-}$ about 395 kJ mol^{-1}. Fe: Z = 26.

Answer Q55–58 on the basis of the above information.

Q55. $[Fe(H_2O)_6]^{2+}$ is: [1]
(A) low spin, $t_{2g}^6e_g^0$
(B) low spin, $t_{2g}^3e_g^3$
(C) high spin, $t_{2g}^4e_g^2$
(D) high spin, $t_{2g}^6e_g^0$

Q56. The number of unpaired electrons in $[Fe(CN)_6]^{4-}$ is: [1]
(A) 2 **(B)** 0 **(C)** 1 **(D)** 4

Q57. The magnetic moment of $[Fe(H_2O)_6]^{2+}$ is about: [1]
(A) 2.83 BM **(B)** 0 BM **(C)** 5.92 BM **(D)** 4.90 BM

Q58. The hybridisation of Fe in $[Fe(CN)_6]^{4-}$ (VBT) is: [1]
(A) sp^3d^2 **(B)** dsp^2
(C) d^2sp^3 **(D)** sp^3

Section E

5-mark level (Long Answer type) · 2 × 5 = 10 marks · −1.25 per wrong

Q59. Draw the structures of the geometrical isomers of $[Co(NH_3)_4Cl_2]^+$ and $[Co(NH_3)_3Cl_3]$, and state which (if any) are optically active. The correct description is: [5]
Isomerism

(A) $[Co(NH_3)_4Cl_2]^+$: cis and trans; $[Co(NH_3)_3Cl_3]$: only the mer form exists
(B) $[Co(NH_3)_4Cl_2]^+$: cis and trans; $[Co(NH_3)_3Cl_3]$: fac and mer; none is optically active
(C) $[Co(NH_3)_4Cl_2]^+$: cis and trans, cis is optically active; $[Co(NH_3)_3Cl_3]$: fac and mer
(D) $[Co(NH_3)_4Cl_2]^+$: only one form; $[Co(NH_3)_3Cl_3]$: fac and mer, both optically active

Q60. For the complex $[Fe(en)_2Cl_2]Cl$, identify (i) the oxidation number of iron, (ii) the hybrid orbitals and shape of the complex, (iii) its magnetic behaviour (assume en makes it low spin), (iv) the number of its geometrical isomers, (v) whether there is an optical isomer, and (vi) its IUPAC name. [5]
Bonding (VBT & CFT), Colour & Magnetism

(A) (i) +3; (ii) d^2sp^3, octahedral; (iii) paramagnetic, 1 unpaired e^-; (iv) 2; (v) yes, the cis form; (vi) dichloridobis(ethane-1,2-diamine)iron(III) chloride
(B) (i) +3; (ii) dsp^2, square planar; (iii) paramagnetic, 1 unpaired e^-; (iv) 2; (v) yes, the cis form; (vi) dichloridobis(ethane-1,2-diamine)iron(III) chloride
(C) (i) +3; (ii) d^2sp^3, octahedral; (iii) diamagnetic; (iv) 3; (v) no; (vi) bis(ethane-1,2-diamine)dichloridoferrate(III) chloride
(D) (i) +2; (ii) sp^3d^2, octahedral; (iii) paramagnetic, 5 unpaired e^-; (iv) 2; (v) yes, the trans form; (vi) dichloridobis(ethane-1,2-diamine)iron(II) chloride

◆ END OF QUESTION PAPER ◆